

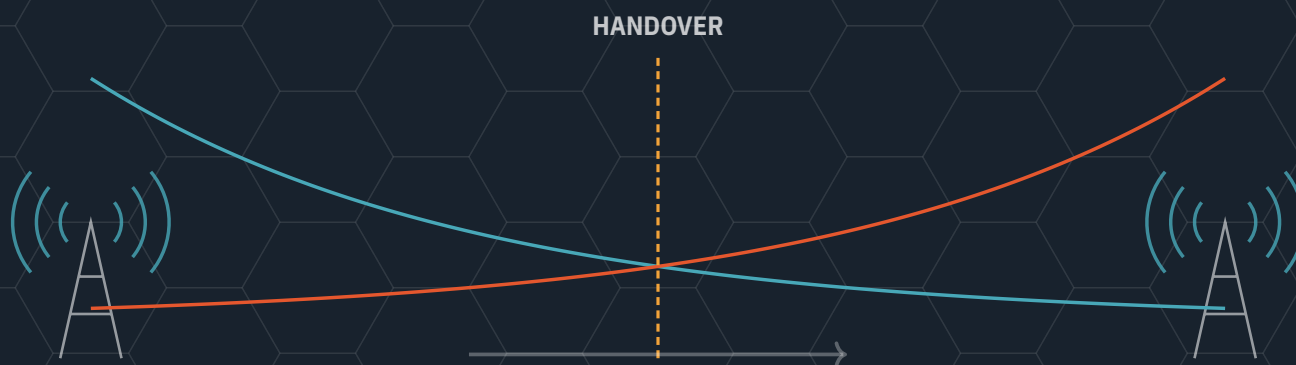
Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 4

Mobility and Handover

تحرک و تحویل



Lecture Notes

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Learning Objectives

By the end of this module, you will be able to:

- Explain WHY mobility management is essential in cellular networks
- Distinguish between Location Areas, Routing Areas, and Tracking Areas
- Describe cell reselection mechanisms in idle mode
- Classify and compare handover types across generations
- Analyze the handover decision process using measurement events
- Trace the LTE X2 handover procedure step by step
- Identify handover failures and their mitigation strategies
- Compare mobility evolution from 2G through 5G

1 Why Mobility Management Is Needed

WHY

Users are mobile — they walk, drive, and travel while expecting uninterrupted voice calls, video streams, and data sessions. Without mobility management, a user crossing a cell boundary would experience immediate service disconnection.

WHAT

Mobility management is the set of network procedures that:

- **Track** the user's location (so the network can reach them for incoming calls/data)
- **Maintain** service continuity when the user moves between cells (handover)
- **Optimize** radio resource usage by connecting users to the best-serving cell

HOW

The network achieves this through two complementary mechanisms:

State	Mechanism	Purpose
Idle mode	Cell reselection + location/tracking updates	Network knows user's approximate location
Connected mode	Handover	Seamless transfer of active connection

The Fundamental Trade-off

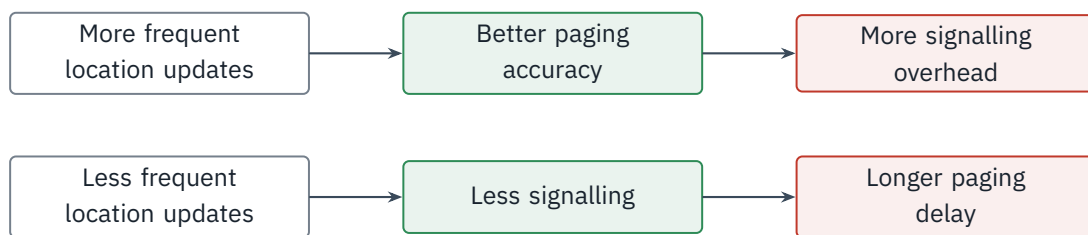


Figure 4.1 The location-update trade-off: paging accuracy versus signalling load

The network must balance:

- **Signaling load** (updates consume radio resources)
- **Paging load** (larger areas = more cells to page)
- **Service latency** (time to reach the user for incoming services)

2 Location Management: LA vs RA vs TA

WHY

The network needs to know **WHERE** a user is (approximately) to deliver incoming calls or data. Different generations use different grouping strategies to balance signaling vs. paging efficiency.

WHAT

Generation	Area Concept	Used For	Typical Size
2G (GSM)	Location Area (LA)	Circuit-switched services	20–100 cells
2.5G (GPRS)	Routing Area (RA)	Packet-switched services	Subset of LA, 10–50 cells
4G (LTE)	Tracking Area (TA)	All services (all-IP)	30–80 cells
5G (NR)	Tracking Area + RNA	All services + RAN-level	Configurable per UE

HOW – Location Area (2G)

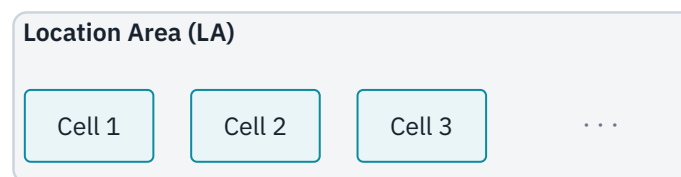


Figure 4.2 A Location Area groups many cells

- UE performs **Location Area Update (LAU)** when crossing LA boundary
- Network pages ALL cells in the LA for incoming call
- Managed by MSC/VLR

HOW – Routing Area (GPRS)

- RA is a **subset** of an LA (one LA contains multiple RAs)
- UE performs **Routing Area Update (RAU)** for packet services
- Managed by SGSN
- Allows finer-grained location for data users

HOW – Tracking Area (LTE/5G)

- **Tracking Area List (TAL):** UE is assigned a LIST of TAs — no update needed when moving within the list
- Reduces signaling for high-mobility users (larger TAL)
- Low-mobility users get smaller TAL to reduce paging area

```
UE TAL = {TA1, TA2, TA3, TA4}
```

```
Moving TA1→TA2→TA3→TA4: NO update needed
```

```
Moving TA4→TA5 (outside TAL): Tracking Area Update triggered
```

■ 5G Enhancement: RAN-based Notification Area (RNA)

- In **RRC_INACTIVE** state (new in 5G), the gNB manages a smaller RNA
- Reduces core network signaling
- Two-level mobility: RNA (RAN-managed) + TA (core-managed)

3 Cell Reselection (Idle Mode Mobility)

■ WHY

Even when not in an active call/session, the UE must camp on the best cell to:

- Receive paging messages efficiently
- Ensure good signal quality when a connection is needed
- Avoid unnecessary location updates

■ WHAT

Cell reselection is the **UE-autonomous** process of selecting a new cell to camp on while in idle mode. The network provides parameters via System Information Broadcasts (SIBs), but the UE makes the decision independently.

■ HOW – Cell Reselection Criteria

■ Ranking Criterion (S-criterion)

A cell is suitable for camping if it meets the S-criterion:

$$S_{\text{rxlev}} = Q_{\text{rxlevmeas}} - (Q_{\text{rxlevmin}} + Q_{\text{rxlevminoffset}}) - P_{\text{compensation}} > 0$$

$$S_{\text{qual}} = Q_{\text{qualmeas}} - (Q_{\text{qualmin}} + Q_{\text{qualminoffset}}) > 0$$

Where:

- $Q_{\text{rxlevmeas}}$: Measured RSRP (dBm)
- Q_{rxlevmin} : Minimum required RSRP (-124 to -44 dBm, typical: -124 dBm)
- Q_{qualmeas} : Measured RSRQ (dB)
- Q_{qualmin} : Minimum required RSRQ (-20 to -3 dB, typical: -20 dB)

■ Cell Ranking (R-criterion)

For intra-frequency reselection, cells are ranked:

$$R_s = Q_{\text{meas},s} + Q_{\text{Hyst}} \quad (\text{serving cell})$$

$$R_n = Q_{\text{meas},n} - Q_{\text{offset}} \quad (\text{neighbour cell})$$

Reselect to the neighbour if $R_n > R_s$ holds for the duration $T_{\text{reselection}}$.

■ Priority-Based Reselection (Inter-frequency/Inter-RAT)

Priority	Action
Higher priority layer	Reselect if neighbor meets $\text{ThreshX}_{\text{High}}$
Equal priority layer	Use ranking (R-criterion)
Lower priority layer	Reselect only if serving < $\text{ThreshServing}_{\text{Low}}$ AND neighbor > $\text{ThreshX}_{\text{Low}}$

■ HOW – Hysteresis and Timers

Hysteresis (QHyst) prevents ping-pong reselection:

- Typical values: 2–6 dB
- Adds “stickiness” to the serving cell

Treselection timer (typically 1–4 seconds):

- Neighbor must remain better for this duration before reselection occurs
- Prevents reselection due to momentary fading

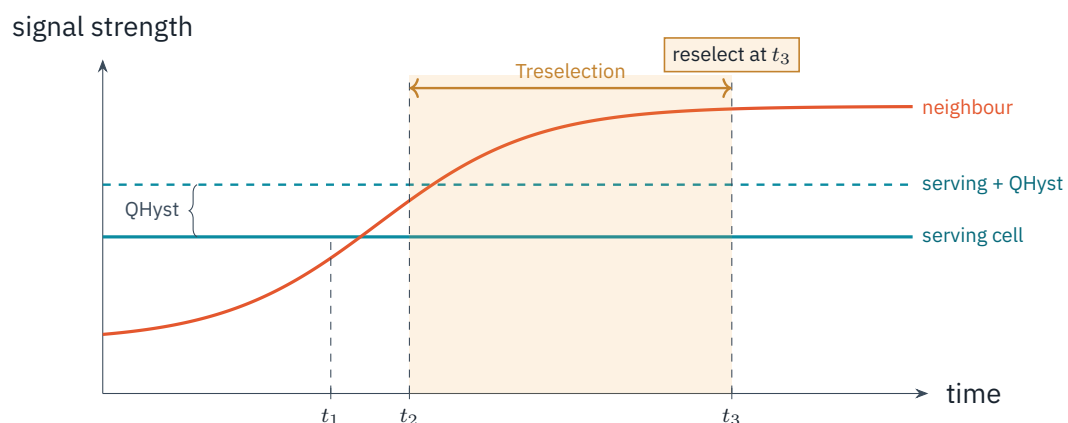


Figure 4.3 Cell reselection: the neighbour must stay above serving + QHyst for Treselection

Typical Cell Reselection Parameters

Parameter	Typical Value	Purpose
QHyst	4 dB	Serving cell hysteresis
Treselection	2 s	Stability timer
Qrxlevmin	-124 dBm	Minimum RSRP for cell selection
Qqualmin	-20 dB	Minimum RSRQ for cell selection
ThreshServing,Low	4 (= 4 dB)	Threshold to leave serving (to lower priority)
ThreshX,High	8 (= 8 dB)	Threshold to enter higher priority layer

4 Handover Types

WHY

Different mobility scenarios require different handover strategies. A user walking between co-channel cells needs a different mechanism than a user on a high-speed train crossing between LTE and 3G coverage areas.

WHAT – Classification Overview

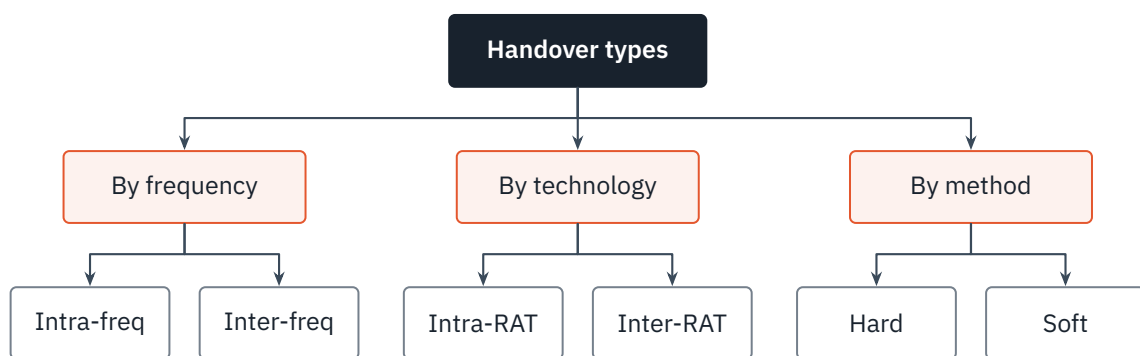


Figure 4.4 Classification of handovers

4.1 Intra-Frequency Handover

■ WHY

Most common handover — occurs when user moves between cells operating on the **same carrier frequency**. This is the typical scenario in dense urban deployments.

■ WHAT

- Source and target cells use the same frequency (e.g., both on 1800 MHz Band 3)
- UE can measure neighbors **WITHOUT** measurement gaps
- Fastest and simplest handover type

■ HOW

- UE continuously measures RSRP/RSRQ of serving and intra-frequency neighbors
- No interruption to measurements needed (same frequency = can measure while receiving)
- Triggered by Event A3: Neighbor becomes offset better than serving

Typical scenario: Walking between adjacent cells in a shopping mall (all cells on same frequency)

4.2 Inter-Frequency Handover

■ WHY

Networks deploy multiple frequency layers for capacity (e.g., macro on 800 MHz for coverage, small cells on 2600 MHz for capacity). Users must move between these layers.

■ WHAT

- Source and target cells operate on **different carrier frequencies**
- UE needs **measurement gaps** to tune its receiver to other frequencies
- Slightly longer interruption than intra-frequency

■ HOW

- Network configures measurement gaps (6 ms every 40 or 80 ms)
- UE measures inter-frequency neighbors during gaps
- Triggered by Event A4 (neighbor on other frequency becomes better than threshold) or A5 (serving degrades AND neighbor is good)

Typical scenario: User moves from indoor small cell (2600 MHz) to outdoor macro (800 MHz)

4.3 Inter-RAT Handover

■ WHY

Not all areas have LTE/5G coverage. When LTE signal degrades, the UE may need to fall back to 3G (UMTS) or even 2G (GSM) to maintain service. Also used for CS Fallback (voice calls).

■ WHAT

- Handover between different radio access technologies
- Examples: LTE → UMTS, LTE → GSM, NR → LTE
- Most complex type — different protocols, different core networks

■ HOW

- UE measures inter-RAT neighbors during measurement gaps
- Triggered by Event B1 (inter-RAT neighbor above threshold) or B2 (serving below threshold AND inter-RAT above threshold)
- Requires inter-RAT coordination between eNB and RNC/BSC

Typical scenario:

- VoLTE not available → CS Fallback to 3G for voice call
- User drives into rural area with only 3G coverage

4.4 Hard Handover (Break-Before-Make)

■ WHY

When simultaneous connection to two cells is not possible (different frequencies, different technologies, or system design choice), the old connection must be released before the new one is established.

■ WHAT

- Connection to source cell is **released** before connecting to target
- Brief interruption in service (typically 20–50 ms in LTE)
- Used in: GSM, LTE, 5G NR

■ HOW

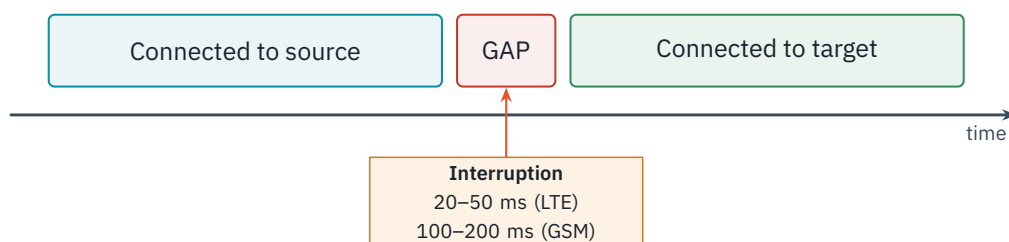


Figure 4.5 Hard handover: break before make

- **Advantage:** Simpler, no need for combining/splitting
- **Disadvantage:** Brief interruption (acceptable for data, noticeable for voice in 2G)

4.5 Soft Handover (Make-Before-Break) – 3G UMTS Specific

■ WHY

CDMA-based 3G systems can receive the same signal from multiple cells simultaneously (same frequency, code division). This enables seamless handover with NO interruption.

■ WHAT

- UE is connected to **multiple cells simultaneously** (Active Set)
- Signals from multiple cells are **combined** for diversity gain
- Unique to CDMA/WCDMA (3G) — NOT used in LTE or 5G

■ HOW – Active Set Management

Active Set Events (3GPP UMTS):

- Event 1A: Add cell to Active Set (neighbor strong enough)
- Event 1B: Remove cell from Active Set (cell too weak)
- Event 1C: Replace cell in Active Set (better candidate found)

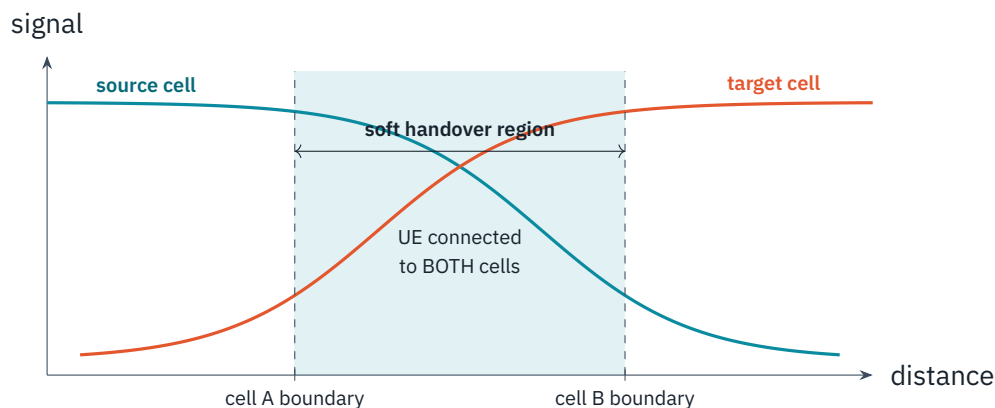


Figure 4.6 Soft handover region: the UE is connected to both cells

- **Active Set:** Set of cells currently serving the UE (max 3–6 cells)
- **Soft handover:** Between cells of same Node B or different Node Bs, same frequency
- **Softer handover:** Between sectors of the SAME Node B (MRC combining at Node B)

Advantages:

- No interruption
- Macro-diversity gain (2–3 dB)
- Reduced ping-pong

Disadvantages:

- Uses resources in multiple cells simultaneously
- Increased backhaul and processing requirements
- Adds complexity to power control

5 Handover Decision Process

WHY

The network must decide WHEN to handover, to WHICH cell, and ensure the decision is robust against fading and noise. A poorly timed handover causes drops; a well-tuned one is imperceptible to the user.

WHAT

The handover decision in LTE is based on:

1. **UE Measurement Reports** — periodic or event-triggered
2. **Measurement Events** — conditions that trigger the report
3. **Hysteresis and Time-to-Trigger** — filtering to avoid false triggers
4. **Network decision** — eNB evaluates report and decides

5.1 Measurement Reports (RSRP and RSRQ)

WHAT

Metric	Full Name	Range	Purpose
RSRP	Reference Signal Received Power	-140 to -44 dBm	Signal strength (coverage)
RSRQ	Reference Signal Received Quality	-20 to -3 dB	Signal quality (accounts for interference + noise)
SINR	Signal-to-Interference-plus-Noise Ratio	-23 to 40 dB	Overall link quality

HOW

RSRP = power of one resource element carrying the reference signal

$$\text{RSRQ} = \frac{N \times \text{RSRP}}{\text{RSSI}}$$

where N is the number of RBs and RSSI is the total received wideband power.

- RSRP answers “how strong is **my** cell’s signal?”
- RSRQ answers “how good is **my** cell’s signal relative to everything else I hear?”

When to use which:

- RSRP-based handover: Good for coverage-limited scenarios (rural)
- RSRQ-based handover: Good for interference-limited scenarios (dense urban)

5.2 LTE Measurement Events (A1–A6)

■ WHAT

Event	Condition	Use Case
A1	Serving > Threshold	Stop inter-freq measurements (serving is good enough)
A2	Serving < Threshold	Start inter-freq/inter-RAT measurements
A3	Neighbor > Serving + Offset	Primary intra-freq handover trigger
A4	Neighbor > Threshold	Inter-frequency handover
A5	Serving < Thresh1 AND Neighbor > Thresh2	Inter-freq/inter-RAT handover
A6	Neighbor > SCell + Offset	Secondary cell (CA) change

■ Event B (Inter-RAT):

Event	Condition	Use Case
B1	Inter-RAT neighbor > Threshold	Redirect to other RAT
B2	Serving < Thresh1 AND Inter-RAT > Thresh2	Fallback (e.g., LTE→3G)

■ HOW – Event A3 in Detail (Most Important)

Event A3 entering condition:

$$M_n + O_{fn} + O_{cn} - H_{ys} > M_s + O_{fs} + O_{cs} + O_{ff}$$

Symbol	Meaning
M_n	Measurement of the neighbour cell (RSRP or RSRQ)
M_s	Measurement of the serving cell
O_{fn}	Frequency-specific offset for the neighbour
O_{fs}	Frequency-specific offset for the serving cell
O_{cn}	Cell-specific offset for the neighbour
O_{cs}	Cell-specific offset for the serving cell
H_{ys}	Hysteresis (positive value)
O_{ff}	A3 offset (the main tunable parameter)

Simplified (same frequency, no cell-specific offsets):

$$RSRP_{neighbour} - Hysteresis > RSRP_{serving} + A3_Offset$$

5.3 Hysteresis and Time-to-Trigger

■ WHY

Radio signals fluctuate due to fading. Without filtering, the network would trigger handovers on every temporary fade, causing excessive signaling and ping-pong.

■ WHAT

Two mechanisms prevent false triggering:

Parameter	Function	Typical Value
Hysteresis (Hys)	Signal must exceed threshold by this margin	1–3 dB
Time-to-Trigger (TTT)	Condition must be sustained for this duration	40–640 ms

■ HOW – Visualization

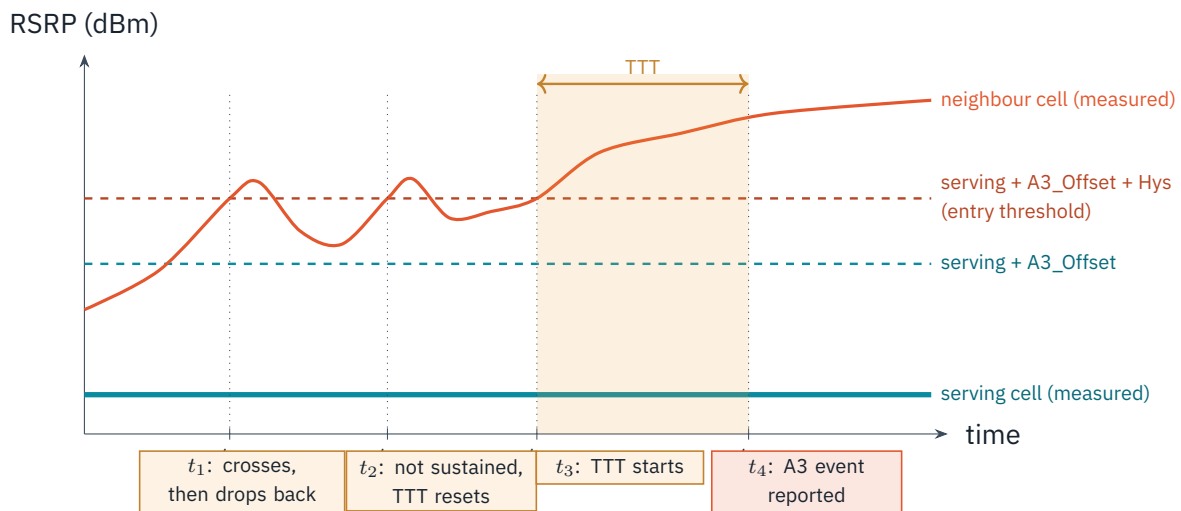


Figure 4.7 Event A3: entry threshold, hysteresis and time-to-trigger

Combined effect:

- Hysteresis provides amplitude filtering (dB domain)
- TTT provides time filtering (time domain)
- Together they provide robust handover triggering

■ Engineering Trade-off

	Low Hys + Short TTT	High Hys + Long TTT
Handover speed	Fast handover	Slow handover
Main risk	Ping-pong	Too-late HO
Good for	Slow users	Fast users
Signalling	More signalling	Less signalling
Typical setting	Urban: Hys = 2 dB, TTT = 160 ms	Highway: Hys = 3 dB, TTT = 80 ms

5.4 Handover Margin

■ WHAT

The **total handover margin** is the combined effect of A3 offset, hysteresis, and TTT that determines how much better a neighbor must be before handover occurs.

$$\text{Effective HO margin} \approx \text{A3_Offset} + \text{Hysteresis} + \underbrace{(\text{UE speed} \times \text{TTT})}_{\text{signal change during TTT}}$$

■ Example Calculation

Given:

A3 Offset = 3 dB

Hysteresis = 2 dB

TTT = 320 ms

UE speed = 50 km/h → path loss changes ~0.5 dB during TTT

Effective margin = 3 + 2 + 0.5 = 5.5 dB

Meaning: Neighbor must be 5.5 dB better than serving
before handover is triggered

6 Handover Procedure – LTE X2 Handover

■ WHY

The X2-based handover is the **preferred** handover mechanism in LTE because it enables direct eNB-to-eNB communication without routing through the core network (MME), reducing latency and signaling load.

■ WHAT

- Flat architecture: eNBs communicate directly via X2 interface
- UE-assisted, network-controlled: UE measures, eNB decides
- Preparation phase ensures target has resources BEFORE handover command
- Total interruption time: **20–50 ms** (imperceptible for most applications)

■ HOW – X2 Handover Sequence

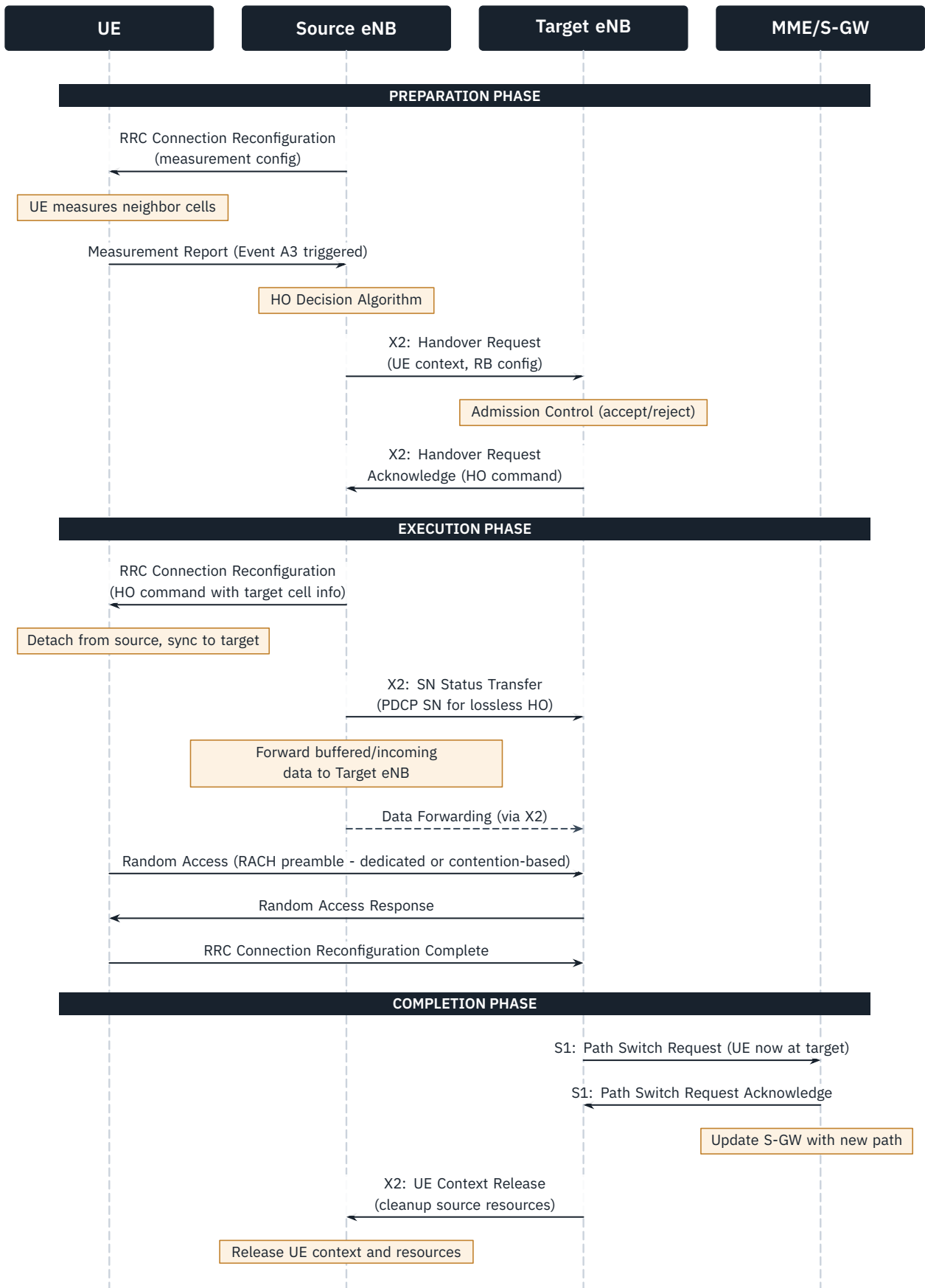


Figure 4.8 LTE X2 handover: preparation, execution and completion

■ Phase-by-Phase Breakdown

■ Phase 1: Preparation (~50–200 ms)

Step	Message	Purpose
1	Measurement Report	UE informs source about neighbor quality
2	HO Decision	Source eNB runs algorithm based on report
3	Handover Request (X2)	Source asks target to prepare resources
4	Admission Control	Target checks if it can accept the UE
5	HO Request Ack (X2)	Target confirms, provides HO command

■ Phase 2: Execution (~20–50 ms interruption)

Step	Message	Purpose
6	RRC Reconfiguration	Source commands UE to handover
7	SN Status Transfer	Ensures no data loss (PDCP SN alignment)
8	Data Forwarding	Buffered data forwarded to target
9	Random Access	UE synchronizes to target cell
10	Reconfig Complete	UE confirms successful handover

■ Phase 3: Completion (~20–100 ms)

Step	Message	Purpose
11	Path Switch Request	Informs MME/S-GW of new location
12	Path Switch Ack	Core confirms path update
13	UE Context Release	Source releases resources

■ Key Timing Values

Parameter	Typical Value
Total preparation time	50–200 ms
User-plane interruption	20–50 ms
Path switch completion	20–100 ms
Total end-to-end	100–350 ms
RACH preamble	Dedicated (no contention) or contention-based

■ S1-based Handover (Fallback)

When X2 interface is NOT available (no direct connection between eNBs):

- All messages route through MME
- Longer preparation time (~100–500 ms)
- Used for inter-eNB handover without X2, or inter-MME handover
- Same UE experience but higher network latency

7 Handover Failures

■ WHY

Handover failures directly impact user experience — dropped calls, interrupted video, lost data. Understanding failure modes is essential for network optimization.

■ WHAT

Four primary failure modes exist, each with distinct causes and solutions:

7.1 Too-Late Handover

■ WHY it happens

The UE moves too fast or the handover margin is too large, so the serving cell signal drops below usable level BEFORE the handover completes.

■ WHAT happens

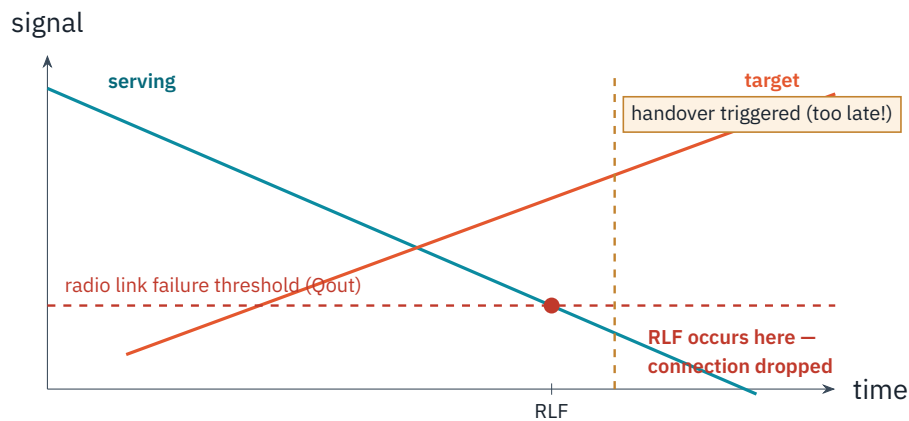


Figure 4.9 Too-late handover: the serving link fails before the handover

■ Indicators

- Radio Link Failure (RLF) in source cell BEFORE HO command received
- High RLF counter on source cell
- RLF report shows strong neighbor at time of failure

■ Mitigation

- **Decrease** A3 offset (trigger HO earlier)
- **Decrease** TTT (react faster)
- **Decrease** hysteresis
- Add early measurement reporting (A2 event with lower threshold)

7.2 Too-Early Handover

■ WHY it happens

Handover triggers too quickly (parameters too aggressive). UE is handed to a cell that is not yet strong enough to sustain the connection.

■ WHAT happens

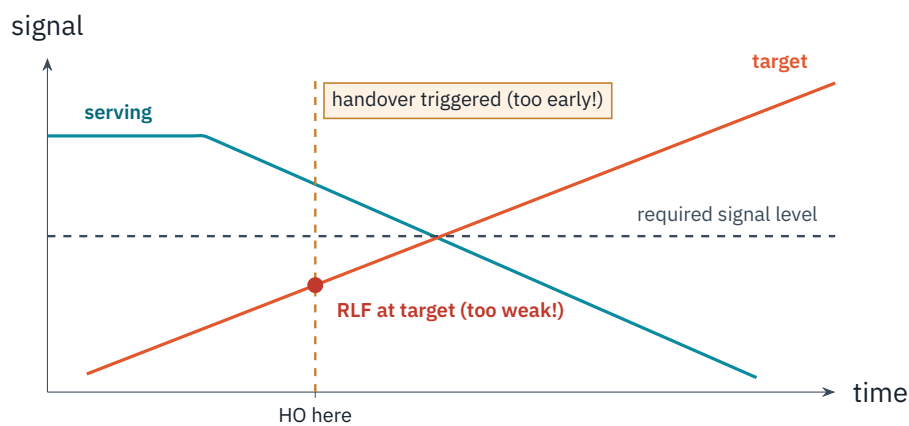


Figure 4.10 Too-early handover: the target cell cannot yet sustain the link

■ Indicators

- RLF occurs at TARGET cell shortly after handover completion
- UE attempts re-establishment back at source cell
- Short time between HO completion and RLF

■ Mitigation

- **Increase** A3 offset (require bigger difference)
- **Increase** TTT (wait longer)
- **Increase** hysteresis
- Verify neighbor cell coverage footprint

7.3 Ping-Pong Effect

■ WHY it happens

Two cells have similar signal strength in an overlap area. With aggressive handover parameters, the UE bounces repeatedly between them.

■ WHAT happens

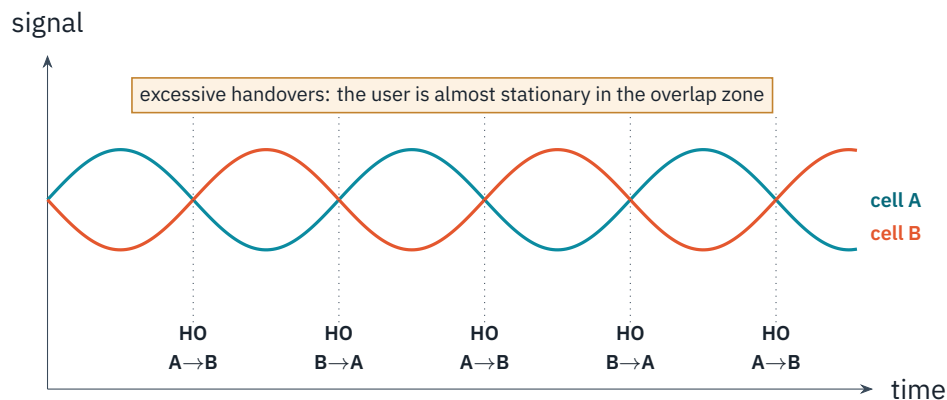


Figure 4.11 Ping-pong: repeated handovers between two similar cells

■ Impact

- Excessive signaling load
- Interrupted user data (each HO = 20-50 ms break)
- Wasted radio resources for HO preparation
- Poor QoE for the user

■ Indicators

- Same UE performing multiple handovers between same cell pair within short time
- High handover rate with low mobility
- A→B followed by B→A within seconds

■ Mitigation

Strategy	How it helps
Increase hysteresis	Larger margin needed to trigger → fewer triggers
Increase TTT	Must sustain longer → transient fluctuations filtered
Cell-Individual Offset (CIO)	Bias one cell to be preferred in overlap
Increase A3 offset	Neighbor must be significantly better
Antenna tilt optimization	Reduce overlap zone size

Rule of thumb: If >30% of handovers between a cell pair are ping-pong (HO back within 5s), increase TTT or hysteresis for that pair.

7.4 Handover to Wrong Cell

■ WHY it happens

The best target cell is not the one selected. This can occur due to:

- Missing neighbor relation (target not in neighbor list)
- Measurement report doesn't include the actual best cell
- Propagation anomaly (signal from far cell momentarily strong)

■ WHAT happens

- UE is handed to Cell X, but Cell Y would have been the correct target
- UE quickly experiences poor signal at Cell X
- May trigger another handover (to Cell Y) or RLF

■ Mitigation

- **Automatic Neighbor Relation (ANR):** Automatically discover and add neighbors
- **Comprehensive neighbor lists:** Ensure all potential targets are configured
- **Cell-specific offsets:** Deprioritize cells known to cause issues
- **Layer-3 filtering:** Average measurements to reduce anomalies

7.5 Handover Failure State Diagram

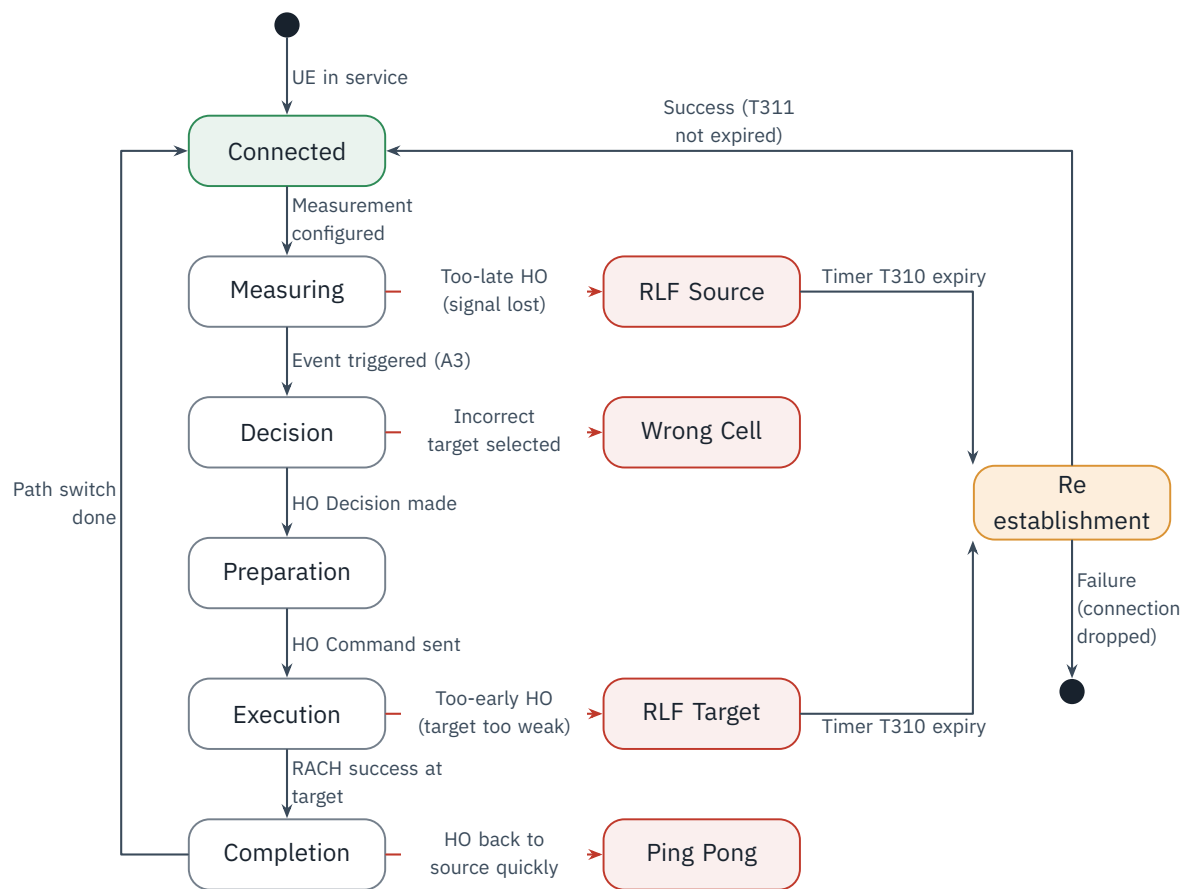


Figure 4.12 Handover states and failure modes

8 Evolution of Handover: 2G → 3G → LTE → 5G

WHY

Each generation improved handover mechanisms to support higher speeds, lower latency, and more complex network architectures. Understanding evolution reveals the design rationale.

8.1 GSM (2G): Hard Handover, BSC-Controlled

Architecture

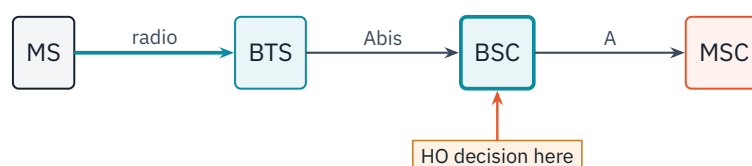


Figure 4.13 GSM: hard handover decided by the BSC

■ Characteristics

Aspect	2G GSM
Handover type	Hard only (break-before-make)
Decision maker	BSC (Base Station Controller)
Measurements	MS measures up to 6 neighbors (BCCH carriers)
Measurement rate	Every SACCH period (480 ms)
Interruption	100–200 ms
Criteria	RXLEV (signal strength), RXQUAL (BER-based quality)
Power control	Combined with handover decision

■ How it worked

1. MS continuously measures serving + 6 strongest neighbors (from BA list)
2. MS reports measurements every 480 ms via SACCH
3. BSC applies averaging (HREQAVE, HREQT parameters)
4. BSC decides based on multiple criteria:
 - Power budget (PBGT) → coverage-based
 - RXQUAL → quality-based (interference)
 - RXLEV → level-based (signal too weak/strong)
 - Distance → timing advance too large
 - Traffic → congestion relief

■ Limitations

- Hard handover: audible click on voice calls
- BSC-controlled: all intelligence in network, slow adaptation
- Limited measurements: only 6 neighbors, slow reporting
- No soft handover: no diversity gain at cell edge

8.2 UMTS (3G): Soft Handover, RNC-Controlled

■ Architecture

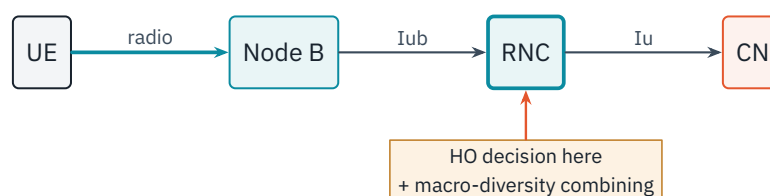


Figure 4.14 UMTS: handover decided by the RNC

■ Characteristics

Aspect	3G UMTS
Handover type	Soft + Softer + Hard
Decision maker	RNC (Radio Network Controller)
Measurements	CPICH Ec/No, RSCP
Active Set	Up to 3–6 cells simultaneously
Interruption	0 ms (soft HO), 20–50 ms (hard HO)
Criteria	Events 1A–1C (active set), 2A–2F (inter-freq/RAT)

■ Soft Handover Mechanism

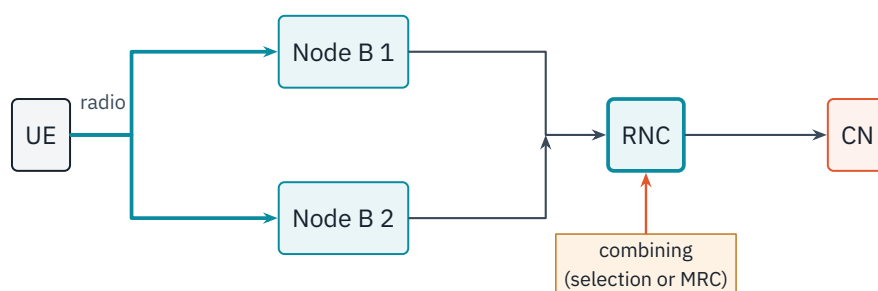


Figure 4.15 Soft handover: two radio legs combined in the RNC

○ Active Set Management:

- Event 1A: $\text{Pilot} > \text{Best_Pilot} - \text{Reporting_Range} + \text{Hysteresis} \rightarrow \text{ADD cell}$
- Event 1B: $\text{Pilot} < \text{Best_Pilot} - \text{Reporting_Range} - \text{Hysteresis} \rightarrow \text{REMOVE cell}$
- Event 1C: $\text{New pilot} > \text{Worst_in_ActiveSet} + \text{Hysteresis} \rightarrow \text{REPLACE cell}$

○ Macro-diversity gain: 2–3 dB at cell edge

○ 20–40% of connections in soft handover in typical network

■ Key Innovation

Soft handover enabled seamless mobility but at the cost of:

- Multiple Channelization Codes consumed (one per active set member)
- Increased Iub backhaul bandwidth
- Complex power control (must satisfy all links)

8.3 LTE (4G): Hard Handover, UE-Assisted Network-Controlled, X2-Based

■ Architecture

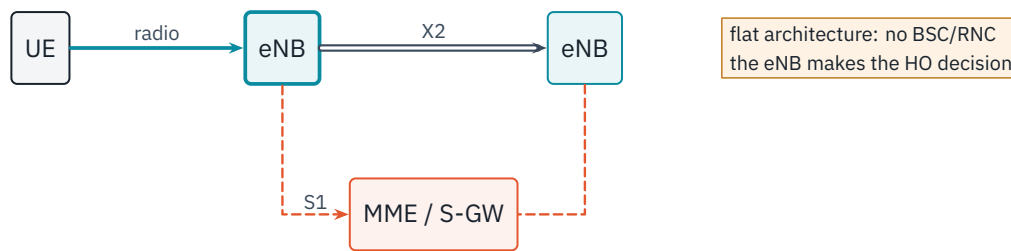


Figure 4.16 LTE: flat architecture, the eNB decides and uses X2

■ Characteristics

Aspect	4G LTE
Handover type	Hard only (break-before-make)
Decision maker	Source eNB (distributed decision)
Measurements	RSRP, RSRQ (events A1–A6, B1–B2)
Interruption	20–50 ms
Interface	X2 (direct eNB-to-eNB)
Key feature	UE-assisted, network-controlled

■ Why LTE Dropped Soft Handover

1. **OFDMA** (not CDMA): Different cells use different subcarriers → cannot combine same signal from multiple cells
2. **Flat architecture**: No central RNC to perform combining
3. **Hard HO is fast enough**: 20–50 ms interruption is acceptable for data
4. **Simplicity**: Significantly simpler than managing Active Sets
5. **Efficiency**: No wasted resources for multiple simultaneous links

■ Key Innovations in LTE

- **X2 interface**: Direct eNB-eNB → fast preparation, data forwarding
- **Event-based reporting**: Only report when conditions met (less signaling)
- **Layer-3 filtering**: Configurable averaging to smooth measurements
- **Automatic Neighbor Relations (ANR)**: Self-configuring neighbor lists
- **Mobility Robustness Optimization (MRO)**: Self-healing handover parameters

8.4 5G NR: Conditional Handover, DAPS, Dual Connectivity

■ Architecture

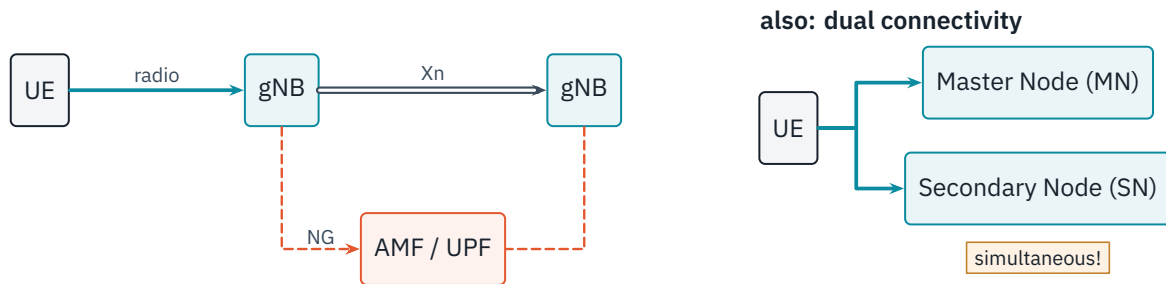


Figure 4.17 5G NR: Xn handover and dual connectivity

■ Characteristics

Aspect	5G NR
Handover type	Hard, Conditional , DAPS
Decision maker	Source gNB (+ CPC for conditional)
Measurements	SS-RSRP, SS-RSRQ, SS-SINR, CSI-RS based
Interruption	0 ms (DAPS), ~0 ms (DC), 20 ms (hard)
Key features	CHO, DAPS, Dual Connectivity, MR-DC

■ Conditional Handover (CHO) – 5G Innovation

WHY: In legacy LTE, the eNB must wait for measurement report, make decision, prepare target, then send command. If radio conditions degrade quickly (high speed, mmWave), this is too slow.

WHAT: Network **pre-prepares** multiple target cells and gives the UE conditions to trigger the handover **autonomously**.

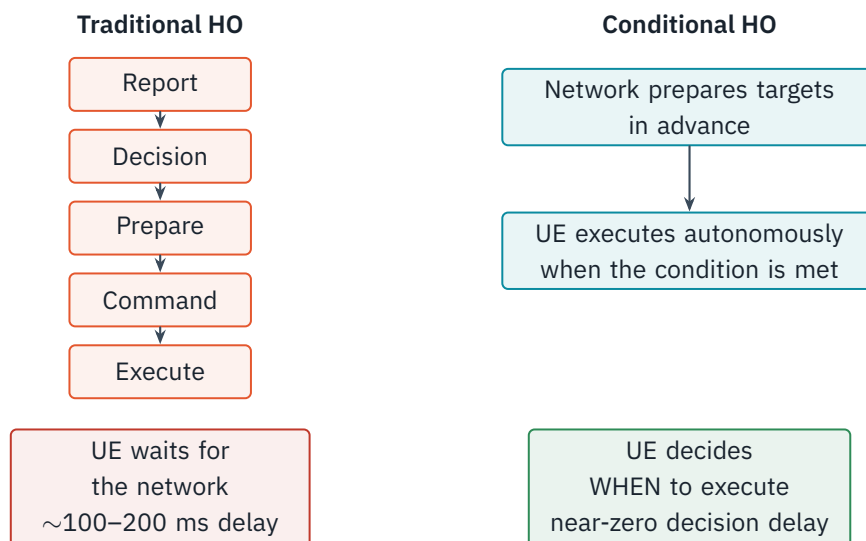


Figure 4.18 Traditional versus conditional handover

HOW:

1. Network configures CHO: “If Event A3 for Cell X OR Cell Y, execute HO to that cell”
2. Target cells are pre-prepared (resources reserved)
3. UE monitors conditions
4. When condition met → UE immediately executes (no round-trip to network)
5. **Benefit:** Eliminates handover preparation delay → fewer failures at high speed

■ Dual Active Protocol Stack (DAPS) – 0 ms Interruption

WHY: Even 20 ms interruption matters for URLLC (Ultra-Reliable Low-Latency Communication).

WHAT: UE maintains connection to BOTH source and target simultaneously during handover — similar concept to 3G soft handover but implemented differently.

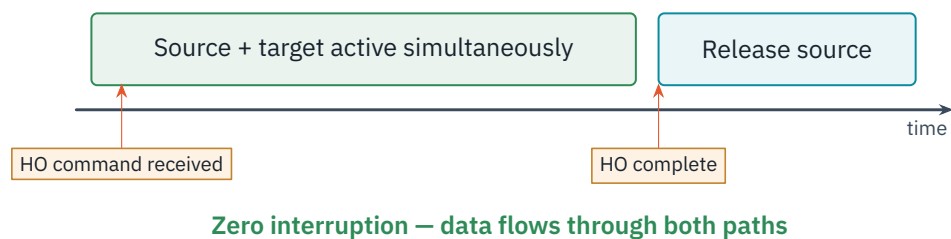


Figure 4.19 DAPS handover: make before break

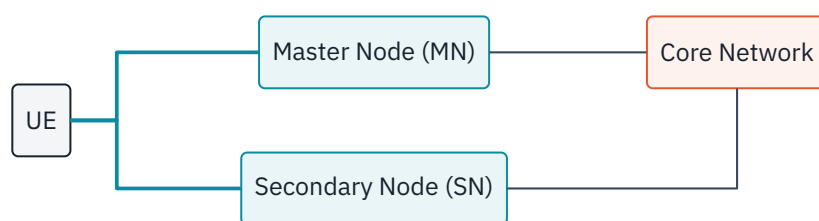
HOW:

- UE has two active protocol stacks (hence the name)
- Data is sent/received through both paths during transition
- Once target confirms successful connection, source is released
- **Trade-off:** Increased UE complexity and power consumption

■ Dual Connectivity (MR-DC) – Multi-RAT

WHAT: UE simultaneously connected to two different nodes:

- **EN-DC** (E-UTRA-NR Dual Connectivity): LTE master + NR secondary
- **NR-DC**: NR master + NR secondary
- **NE-DC**: NR master + LTE secondary



Both links active simultaneously → seamless mobility
SN change ≠ handover (no interruption to MN)

Figure 4.20 Dual connectivity: two radio links at the same time

8.5 Generation Comparison Summary

Feature	2G GSM	3G UMTS	4G LTE	5G NR
HO Type	Hard	Soft + Hard	Hard	Hard + CHO + DAPS
Controller	BSC	RNC	eNB	gNB
Interruption	100-200 ms	0 ms (soft)	20-50 ms	0 ms (DAPS/DC)
Measurement	RXLEV, RXQUAL	CPICH RSCP, Ec/No	RSRP, RSRQ	SS-RSRP, CSI-RS
Intelligence	Network-only	Network-only	UE-assisted	UE-autonomous (CHO)
Architecture	Hierarchical	Hierarchical	Flat	Flat + DC
Self-optimization	Manual	Manual	SON (MRO)	AI/ML-based

9 Engineering Examples with Typical Parameter Values

9.1 Example: Urban Macro Network – LTE Handover Optimization

■ Scenario

- Dense urban area, 500m inter-site distance
- Users: mix of pedestrians (3 km/h) and vehicles (50 km/h)
- Problem: 5% handover failure rate, excessive ping-pong

■ Initial Parameters (before optimization)

Parameter	Value	Issue
A3 Offset	2 dB	Too aggressive → too-early HO
Hysteresis	1 dB	Not enough filtering
TTT	100 ms	Too short → ping-pong
Layer-3 filter (k)	1	Not enough averaging

■ Optimized Parameters

Parameter	New Value	Rationale
A3 Offset	3 dB	More margin before triggering
Hysteresis	2 dB	Better fading resistance
TTT	256 ms	Eliminates most ping-pong for pedestrians
Layer-3 filter (k)	4	Smooths fast fading (~200 ms averaging)
CIO (cell pair with ping-pong)	+3 dB	Biases HO for known problem pair

■ Result

- Handover failure rate: 5% → 1.5%
- Ping-pong rate: 12% → 3%
- Trade-off: Mean HO delay increased by ~150 ms (acceptable)

9.2 Example: High-Speed Rail – 350 km/h

■ Challenge

At 350 km/h, the UE crosses a cell (1 km coverage) in ~10 seconds. Standard parameters are too slow.

■ Parameter Tuning

Parameter	Standard	High-Speed
A3 Offset	3 dB	2 dB
Hysteresis	2 dB	1 dB
TTT	256 ms	40 ms
Layer-3 filter (k)	4	1
A2 threshold	-120 dBm	-110 dBm

■ Additional Measures

- Dedicated RF planning along rail corridor
- Directional antennas pointing along track
- Cell overlap zone: 200–300 m minimum
- Consider **Conditional Handover (5G)** for this scenario

■ Speed Sensitivity Calculation

```

At 350 km/h = 97 m/s
Path loss change rate ≈ 0.3 dB per 10 ms (at 2 GHz, Hata model)

With TTT = 256 ms: Signal changes by ~7.5 dB during TTT!
→ This is unacceptable – must reduce TTT

With TTT = 40 ms: Signal changes by ~1.2 dB during TTT
→ Acceptable margin
    
```

9.3 Example: Inter-RAT Handover – LTE to 3G CSFB

■ Scenario

VoLTE not deployed; voice calls require CS Fallback to 3G.

■ Configuration

Parameter	Value	Purpose
B2 Threshold1 (serving LTE)	-110 dBm RSRP	Trigger when LTE weak
B2 Threshold2 (UMTS target)	-95 dBm RSCP	Only if 3G is adequate
TTT (B2)	640 ms	Avoid unnecessary fallback
A2 Threshold (start measuring 3G)	-105 dBm RSRP	Begin inter-RAT measurements
IRAT measurement gap	6 ms / 40 ms	Gap pattern for measuring UMTS

■ Procedure

1. UE on LTE, RSRP drops below -105 dBm → A2 event → start measuring UMTS
2. LTE RSRP drops below -110 dBm AND UMTS RSCP > -95 dBm for 640 ms → B2 event
3. eNB triggers inter-RAT handover to UMTS
4. UE performs hard handover to 3G, establishes CS domain connection
5. Voice call proceeds on 3G circuit-switched
6. After call ends: UE returns to LTE (fast return / redirection)

9.4 Example: Measurement Event Configuration (Complete)

■ Full A3 Event Configuration for Intra-Frequency HO

```
RRC Connection Reconfiguration (measurement config):
{
  measObjectEUTRA: {
    carrierFreq: 1850 (EARFCN for Band 3, 1800 MHz)
    allowedMeasBandwidth: 50 RBs
    neighCellConfig: 01 (partial measurement)
    offsetFreq: 0 dB
  }

  reportConfigEUTRA: {
    triggerType: event
    eventId: A3
    a3-Offset: 3 dB (= 6 in IE, 0.5 dB steps)
    hysteresis: 2 dB (= 4 in IE, 0.5 dB steps)
    timeToTrigger: 256 ms
    triggerQuantity: RSRP
    reportQuantity: both (RSRP + RSRQ)
    maxReportCells: 4
    reportInterval: 480 ms
    reportAmount: infinity
  }

  measIdToAddList: {
    measId: 1
    measObjectId: 1
    reportConfigId: 1
  }

  quantityConfig: {
    filterCoefficientRSRP: 4 (k=4, ~200ms averaging)
    filterCoefficientRSRQ: 4
  }
}
```

■ Layer-3 Filtering Formula

$$F_n = (1 - a) F_{n-1} + a M_n, \quad a = \left(\frac{1}{2}\right)^{k/4}$$

where k is the filterCoefficient (0, 1, 2, ..., 8, ...):

- $k = 0$: $a = 1.0 \rightarrow$ no filtering (raw measurement used)
- $k = 4$: $a = 0.5 \rightarrow$ 50% new + 50% old $\rightarrow$ about 200 ms memory
- $k = 8$: $a = 0.25 \rightarrow$ 25% new + 75% old $\rightarrow$ about 400 ms memory

9.5 Typical Handover KPIs (Operator Benchmarks)

KPI	Target	Good	Poor
Handover Success Rate	>98%	>99%	<95%
Ping-pong Rate	<5%	<2%	>10%
RLF Rate (HO-related)	<1%	<0.5%	>3%
Mean Interruption Time	<50 ms	<30 ms	>100 ms
Too-early HO ratio	<1%	<0.5%	>3%
Too-late HO ratio	<2%	<1%	>5%
Unnecessary HO (ping-pong + wrong cell)	<5%	<3%	>10%

10 Summary Diagrams

■ Measurement Event A3 Triggering – With Hysteresis and TTT

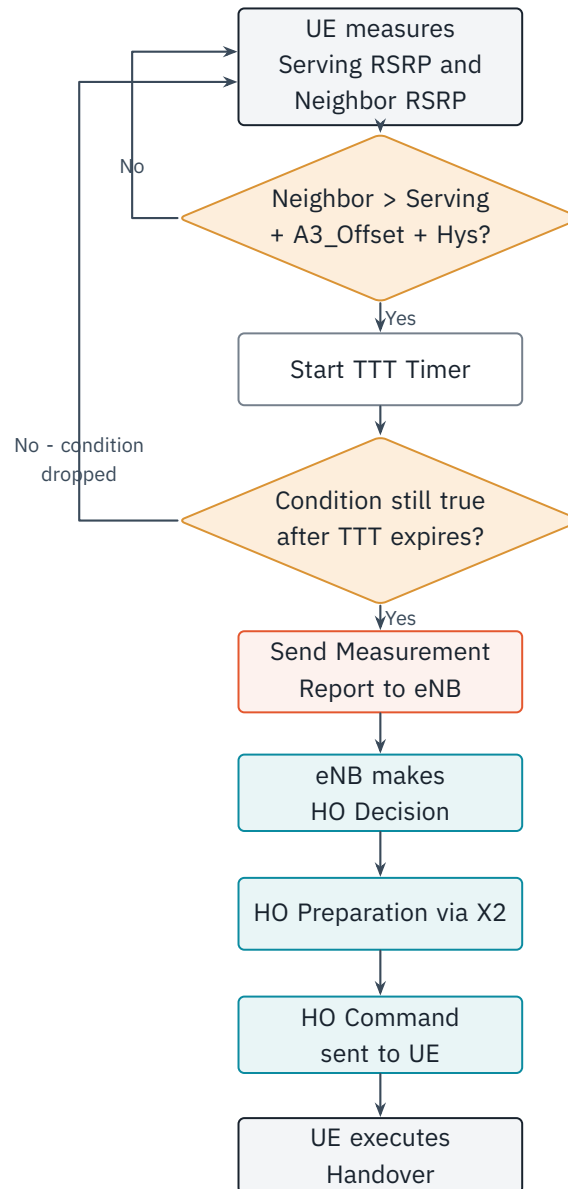


Figure 4.21 Event A3 triggering with hysteresis and time-to-trigger

Handover States – Complete Flow

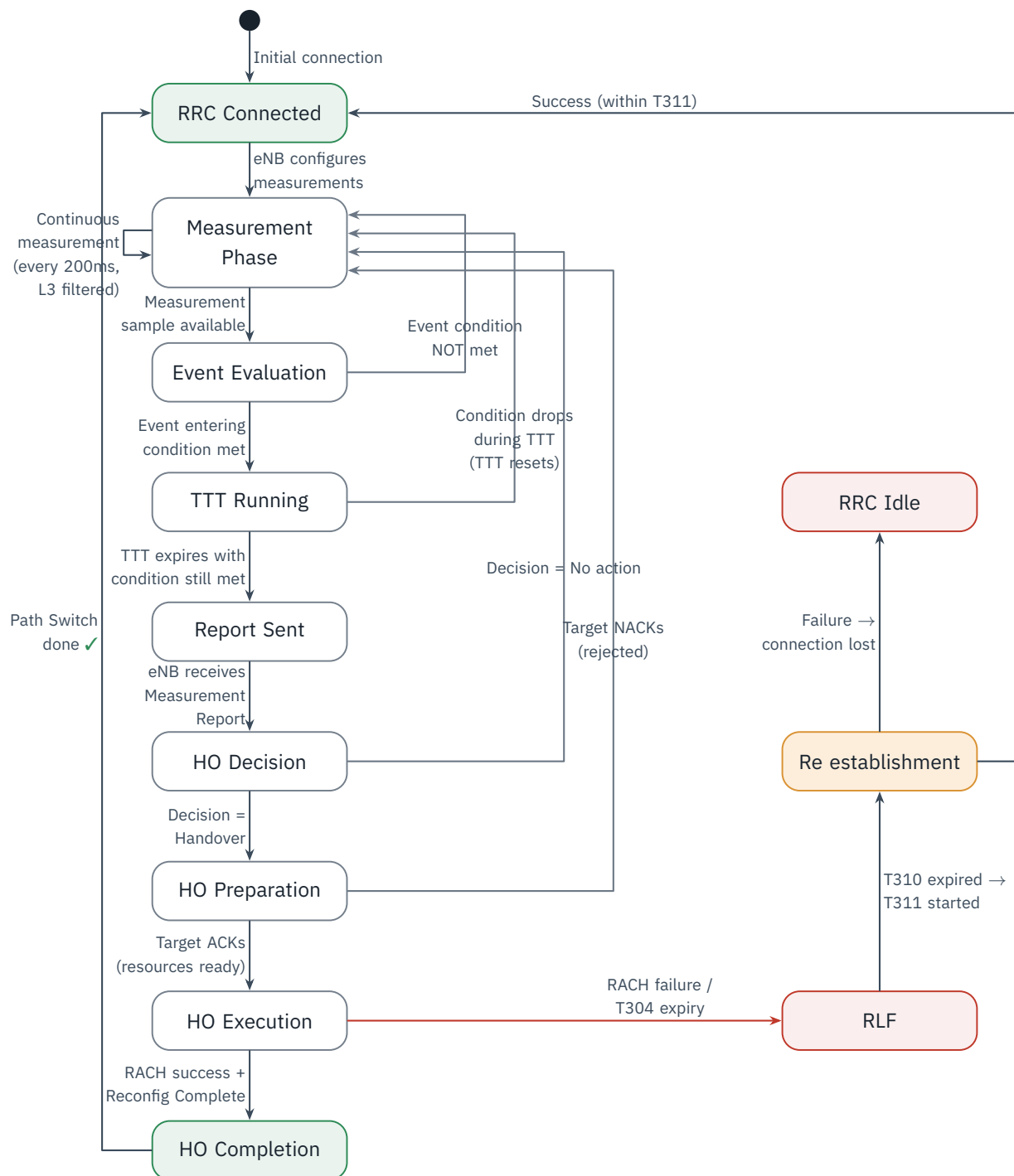


Figure 4.22 Complete handover state flow

LTE Handover Parameters – Decision Tree

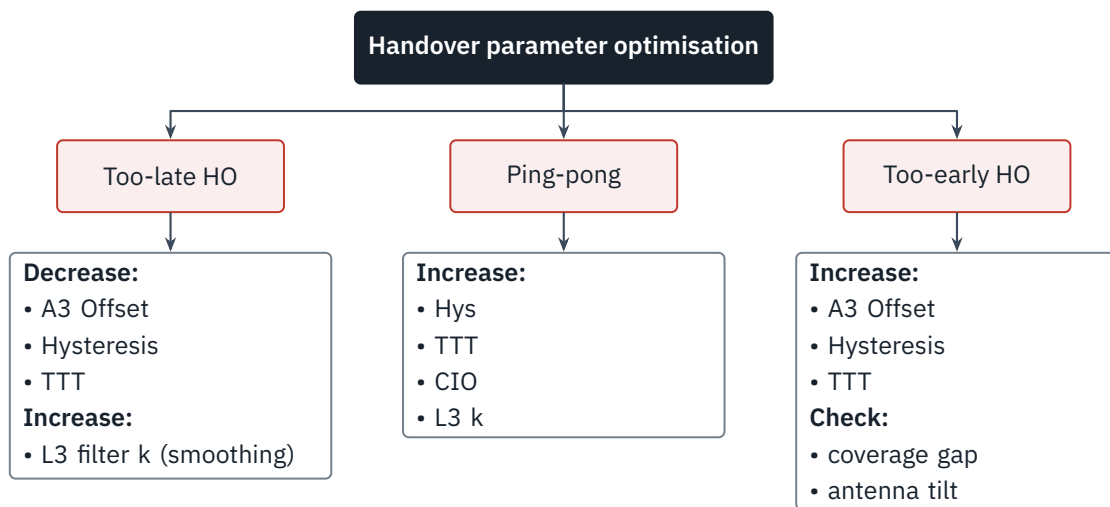


Figure 4.23 Handover parameter optimisation decision tree

Key Formulas Summary

Formula	Expression	Use
A3 Entry	$M_n - Hys > M_s + A3_Off$	Intra-freq HO trigger
A3 Leave	$M_n + Hys < M_s + A3_Off$	Cancel pending report
Cell Ranking (Rs)	$Q_{meas,s} + Q_{Hyst}$	Serving cell reselection rank
Cell Ranking (Rn)	$Q_{meas,n} - Q_{offset}$	Neighbor reselection rank
S-criterion	$S_{rxlev} = Q_{rxlevmeas} - Q_{rxlevmin} > 0$	Cell suitability
L3 Filter	$F_n = (1 - a) F_{n-1} + a M_n$	Measurement smoothing
RSRQ	$N \times RSRP / RSSI$	Signal quality metric
HO Margin	$A3_Off + Hys + speed \times TTT$	Effective trigger margin

Review Questions

1. **Explain** why LTE uses hard handover instead of soft handover like 3G.
2. **Calculate** the effective handover margin for: A3 offset = 4 dB, Hys = 2 dB, TTT = 320 ms, UE speed = 120 km/h.
3. **Draw** the timeline of a too-late handover and identify which parameter to adjust.
4. **Compare** Tracking Areas (LTE) with Location Areas (2G). What is the advantage of Tracking Area Lists?
5. **Design** handover parameters for a highway scenario (130 km/h). Justify each choice.
6. **Explain** how Conditional Handover in 5G solves the high-mobility problem that LTE struggles with.
7. **Analyze** a scenario where ping-pong rate is 15% between two cells. List three mitigation strategies in order of preference.
8. **Describe** the X2 handover procedure and identify where data loss could occur without SN Status Transfer.

Module 4 complete. Next: Module 5 — Radio Resource Management and Scheduling